

Some history

THE GROWTH OF BACTERIOPHAGE

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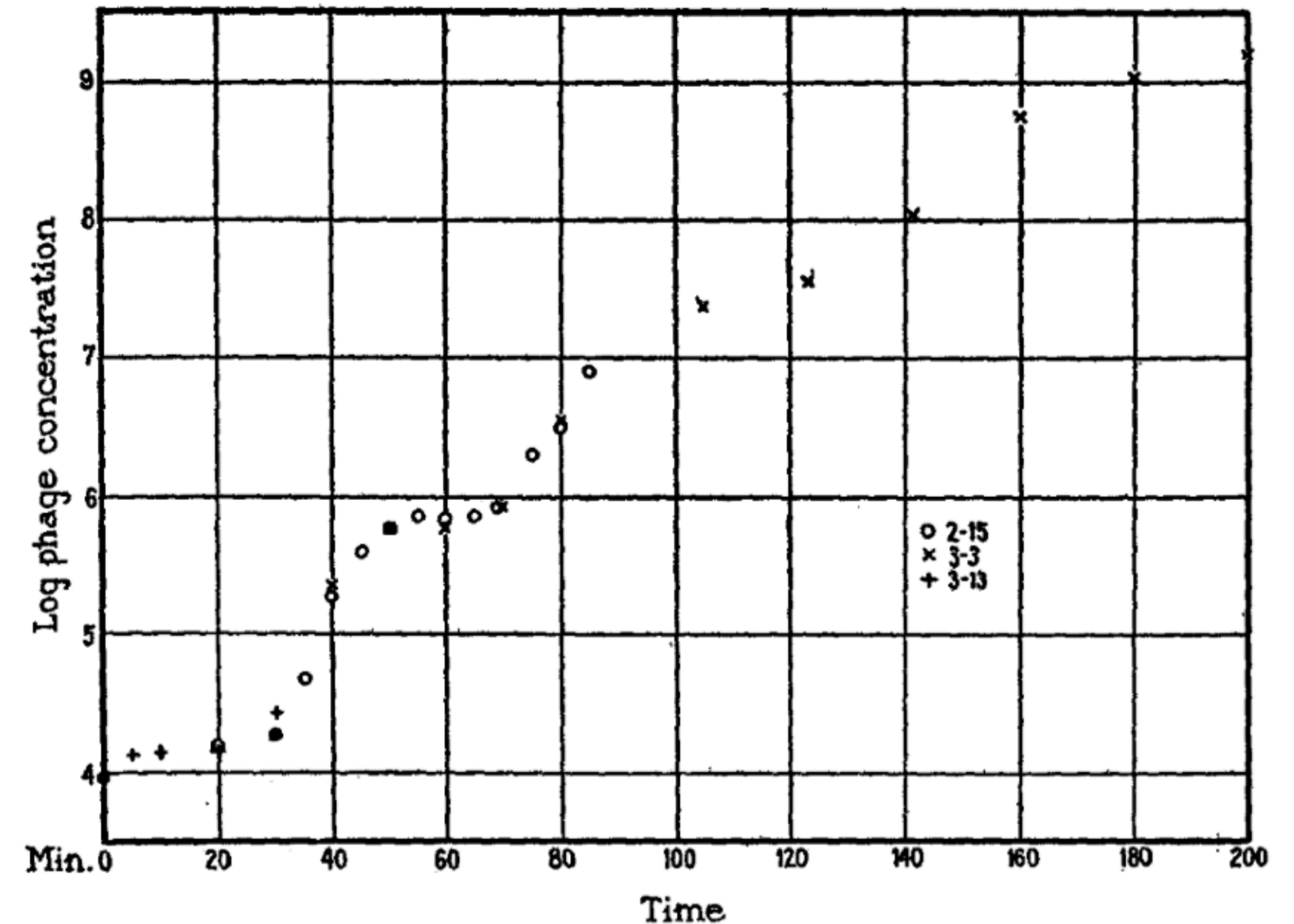
(From the William G. Kerckhoff Laboratories of the Biological Sciences, California
Institute of Technology, Pasadena)

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INTRODUCTION

Certain large protein molecules (viruses) possess the property of multiplying within living organisms. This process, which is at once so foreign to chemistry and so fundamental to biology, is exemplified in the multiplication of bacteriophage in the presence of susceptible bacteria.

Bacteriophage offers a number of advantages for the study of the multiplication process not available with viruses which multiply at the expense of more complex hosts. It can be stored indefinitely in the absence of a host without deterioration. Its concentration can be determined with fair accuracy by several methods, and even the individual particles can be counted by d'Herelle's method. It can be concentrated, purified, and generally handled like nucleoprotein, to which class of substances it apparently belongs (Schlesinger (1) and Northrop (2)). The host organism is easy to culture and in some cases can be grown in purely synthetic media, thus the conditions of growth of the host and of the phage can be controlled and varied in a quantitative and chemically well defined way.



Some history

Quantifying the infectivity of human immunodeficiency virus

(viral infectivity assay/human immunodeficiency virus kinetics/gp120 blocking protein)

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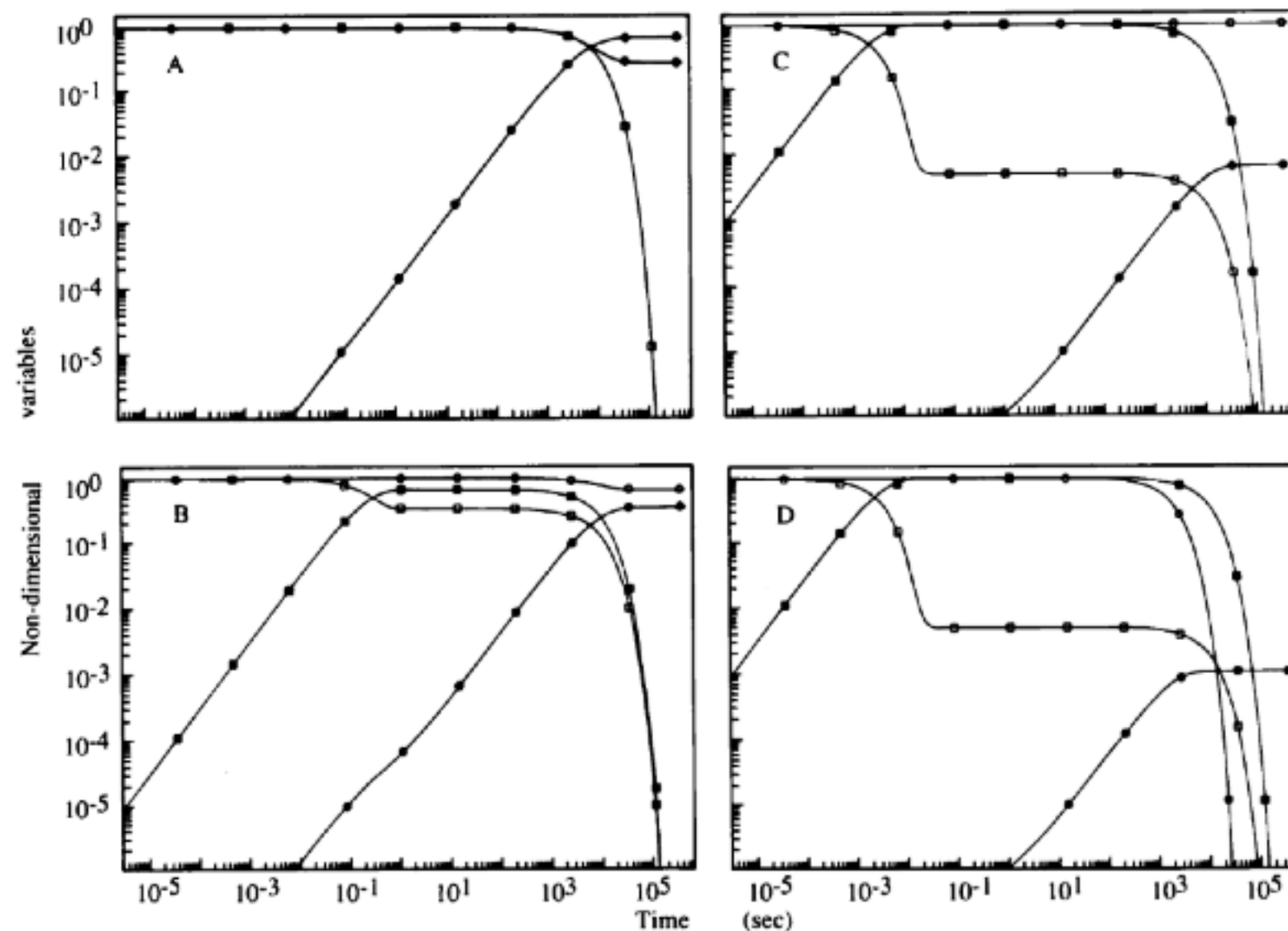
Communicated by Stirling A. Colgate, March 8, 1989 (received for review January 9, 1989)

$$\frac{di}{dt} = N\lambda f v,$$

$$\frac{dv}{dt} = -[N\lambda f + 1 - \sigma]v,$$

$$\frac{df}{dt} = -\gamma(\beta f - c) - \sigma f - \lambda f(1 - f),$$

$$\frac{dc}{dt} = \gamma(\beta f - c) - \sigma c + \lambda f c.$$



Dynamics of HIV Infection of CD4⁺ T cells

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$$\frac{dT}{dt} = s - \mu_T T + rT \left(1 - \frac{T + T^* + T^{**}}{T_{\max}}\right) - k_1 VT, \quad (5a)$$

$$\frac{dT^*}{dt} = k_1 VT - \mu_T T^* - k_2 T^*, \quad (5b)$$

$$\frac{dT^{**}}{dt} = k_2 T^* - \mu_b T^{**}, \quad (5c)$$

$$\frac{dV}{dt} = N\mu_b T^{**} - k_1 VT - \mu_V V. \quad (5d)$$

